PHYSIOLOGY OF THE MOTOR SYSTEM

Have a point

Lecture 03 · writing workshop

The constraints come from outside; the message comes from you

Constraints · from outside

- The question
- The word limit
- The course material

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Message · from you

Your point: what you want the reader to know

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Content · carries the message

- Ideas and facts
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Message · from you

Your point: what you want the reader to know

> The assignment supplies the constraints. Only you can supply the message, and everything else in the answer serves it.

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- “Most important” asks you to choose
- “Why” asks for a reason the reader can follow

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- Room for one point and the evidence it needs

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- Kandel Ch. 35 for the eye; Ch. 34 for the somatic side
- Claims are checked against the chapters
- Pages and figures show where a claim comes from

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> The constraints say what to answer and how much room you have. They do not say what your point is.

Every answer respected the constraints

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- Every answer chose a similarity and a difference
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> The constraints were met. Last week's conversation with the chapter shows in every answer.

The four answers made four different points

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Both systems actively hold still, but the eye can compute its hold from its own command because its mechanics are stable, while the body needs feedback because its loads vary

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Both systems actively hold still, but the eye can compute its hold from its own command because its mechanics are stable, while the body needs feedback because its loads vary

Point D

Both pass a movement through stages from goal to command, but only the eye has a dedicated gate that holds movement back, perhaps because an unwanted eye movement disturbs vision

Other points are possible given the constraints

Point E

Both systems turn a target into a movement, but the eye-movement system is simpler: six muscles for each eye and no load, against many muscles and changing loads

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Point G

Both systems must produce force, but the eye's load never changes, so eye commands specify only position and velocity; arm commands must also compensate for load

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In pairs

- Pick two of these points that you could both write about
- Each of you: say which you prefer, and why

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This answer compares how the eye and the arm relate to vision and to the world

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Ideas put in relationship	The arm moves to change the world. The eye moves to change what we see.

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Not obvious before it is made	The eye-movement system serves vision: we move our eyes in order to see
Ideas put in relationship	The arm moves to change the world. The eye moves to change what we see.
Committed: what you want the reader to accept	In some respects, the eye-movement system could perhaps be seen as partly visual

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One sentence

Both systems use populations of neurons. The eye's lie in the superior colliculus. The arm's lie in several cortical areas.

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One sentence	Both systems use populations of neurons. The eye's lie in the superior colliculus. The arm's lie in several cortical areas.
A claim about the world, not an opinion or a thought	To me, population coding is the most elegant feature of both systems, especially the eye's single map

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Every sentence serves it	… the eye's population lies in one map and the arm's in several cortical areas. The colliculus also has visual layers that map the visual field.

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Interesting to the reader	Both systems must produce force, but eye muscles are smaller than arm muscles
Important to you	Only the writer can check this

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Write several candidates

> “List ten points someone could make about this question, from the obvious to the surprising.”

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Check each	“Which of these are claims that could be wrong, and which would a reader already know?”
Ask what interests you	“Why would someone care whether this point is true?”
Check the evidence	“Which pages and figures in the chapter support this point, and which cut against it?”

Your turn: improve a point

1. Return to your pair.
2. Choose a student point from A–D that neither of you wrote.
3. Improve it as a point.
4. Submit the improved point on Moodle, with the letter of the point you started from.

10 minutes to work.

A few improved points, chosen at random, will be shared with the class.